Xiaoyu Yu (余骁宇)

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Summary

Research interests include **quantitative modeling, machine learning, and data analytics**, **electrocatalytic materials and AI for Science**, and mathematical inverse problems, signal processing, and high-dimensional data reconstruction, with an interest in applications to quantitative finance.

Education

- Ph.D. (Candidate) in Mechanical Engineering., **National University of Singapore** (NUS), Singapore
- M.Sc. in Advancing Theory and Engineering, **Beihang University** (BUAA), China
- B.Sc. (First) in Engineering, **Beihang University** (BUAA), China
- B.Sc. (Second) in Pure and Applied Mathematics, **Beihang University** (BUAA), China

Projects

Mathematical Inverse Modeling and Multichannel Sensor Data Analysis

— Developed acoustic-scattering-based inverse methods to recover dynamic gas–liquid interface information from sensor measurements;
— Independently implemented signal inversion algorithms and custom data-processing software;
— Designed a synchronized multichannel ultrasonic measurement system, achieving millimeter-level bubble measurement accuracy in a cryogenic liquid-nitrogen environment (77 K).

Machine Learning for High-Dimensional Data Reconstruction

— Developed a diffusion-model-based inversion framework using U-Net & EDM, outperforming conventional neural networks with average reconstruction error below 2%.
— Constructed a numerical dataset covering 416 operating conditions using elliptic PDE meshes for model development and evaluation.
— Evaluated reconstruction quality using structural similarity (SSIM) and quantitative error metrics, achieving 0.327% relative error in transient-field reconstruction using DDIM sampling.

Pt-Based Electrocatalyst Engineering for AOR

— Designed and synthesized Pt/NiO-based catalysts by tuning Pt loading, substrates, and thermal treatment to investigate Pt-site configurations and metal–support interactions.
— Integrated CV, LSV, EIS, Tafel, and Raman analyses to assess AOR activity, charge-transfer kinetics, and potential structure–activity relationships associated with surface reconstruction.

Quantitative Imaging and Experimental Data Analysis

— Developed high-speed PIV and 2D3C stereoscopic PIV platforms for quantitative flow-field reconstruction and nanoparticle tracking under extreme conditions (6 MPa, 100–600 m/s).
— Applied high-speed imaging, laser interferometry, and subpixel edge detection to extract high-resolution spatial and temporal measurements of transient physical processes.
— Performed quantitative image and experimental data analysis to characterize particle trajectories, flow dynamics, and small-scale structural vibrations.

Publications & Patents

▷ Journal & Conference Papers (Selected)

[1] Xiaoyu Yu et al. "Impact of vibrating combustion chamber wall on flame thermoacoustic dynamics". In: *Physics of Fluids* 37.11 (Nov. 2025), p. 114111. DOI: 10.1063/5.0302034. URL: https://doi.org/10.1063/5.0302034.

[2] Xiaoyu Yu, Xiaokang Liu, Jingxuan Li, et al. "Theoretical study on intrinsic thermo-acoustic mode in long flame combustion chamber". In: *China National Symposium on Combustion*. China, 2024.

[3] Xiaoyu Yu, Xiaokang Liu, Jingxuan Li, et al. "The Influence of Lateral Offset of Laminar Premixed Flames on Thermoacoustic Instability of Burners". In: *The International Symposium on Multiscale Simulations of Thermophysics*. China, 2024.

▷ Chinese Invention Patents (Selected)

[4] Jingxuan Li, Xiaoyu Yu, et al. Measurement method for liquid bubble content in the test tank and measurement system for liquid bubble content in the test tank: 202410948864X [P] 2024.07.16

[5] Jingxuan Li, Xiaoyu Yu, et al. Adjustable flow solid nanoparticle generator: 2024102667367 [P] 2024.03.08

[6] Jingxuan Li, Xiaoyu Yu, et al. A measurement method and device for gas-liquid interface of liquid jet in a closed environment: 202510096916X [P] 2025.01.22

Work Experience

Student Counselor, Beihang University Student Affairs Office

– Responsible for daily office conference arrangement, administrative management and official document drafting work

– Organized large-scale campus events, award ceremonies and freshman military training activities

Chairman, Graduate Student Union of the School of Astronautics

Conducted comprehensive management and coordination of all functional departments; maintained communication and joint research with institutes and technical units.

COVID-19 Campus Isolation Area Volunteer Leader

Served as person-in-charge of campus isolation zone during pandemic prevention and control period.

Skills, Honors & Scholarships

Programming:	Matlab, LabView, Fortran, C, C++, Python, JavaScript, Origin, Chat-GPT
Software:	AutoCAD Professional Qualification, SolidWorks Professional Qualification, MS Office Level 2 (National Computer Rank Examination)
Languages:	IELTS 6.5, CET-6 565 (Top 15% nationwide), Mandarin (Native), Cantonese (Native)

Selected Scholarships: National Scholarship (2025); BUAA Shenyuan Top Scholarship, CATIC Top-class Scholarship, Multiple First-Class Academic Scholarships; Mathematics & Physics Competition Top Scholarship; Freshman Scholarship.

Selected Honors: National University Student Mathematics Competition First Prize (2021,2022); National Undergraduate Graduation Design Competition Second Prize (2023); BUAA Shenyuan Gold Medal (Top 10 students annually, 2025); Outstanding Graduate; Excellent Student Cadre

Extracurricular Sports: Badminton Team Champion; Football Runner-up (twice); Basketball Team Champion; Table Tennis, Vocal music.